Electrical QA/QC Engineer
Location: Sharjah
Salary: AED 5,000 - 6,000

Industry: Water Treatment

Job Brief:

A leading water treatment company is seeking an Electrical QA/QC Engineer in Sharjah to ensure the quality, compliance, and integrity of all electrical and instrumentation works throughout project execution.

The ideal candidate will have strong experience in QA/QC within water treatment, construction, or oil & gas, along with solid technical knowledge of electrical systems, instrumentation, testing procedures, and quality governance.

Job Responsibilities:

- Accountable for maintaining the quality of every activity related to electrical and instrumentation works, with precise knowledge of all aspects of project quality requirements.
- Manage and maintain QA/QC documentation for the entire project, including certificates, calibration records, test results, inspection requests, non-compliance reports, site instructions or observations, and documentation related to permanent materials delivered to the site.
- Develop and determine quality standards for inspections and testing procedures, oversee testing methods, and ensure adherence to high-quality standards across all processes.

- Review the quality of all materials delivered to the site and ensure compliance with project specifications and quality standards, while coordinating with procurement and engineering teams regarding material quality requirements.
- Supervise the implementation of inspection and testing schedules and ensure adherence to approved procedures and project requirements.
- Coordinate with project teams to perform quality audits on processes and activities across the project lifecycle.
- Ensure employees and team members are aware of quality standards and comply with all quality manuals, procedures, and documentation requirements.
- Collaborate with contractors and suppliers to maintain the quality and compliance of all systems and installations.
- Monitor safe handling, lifting, and storage of equipment and hazardous materials while conducting quality audits as per the required schedule.
- Maintain an efficient monitoring system to record project activities and analyze processes to ensure all work meets defined quality requirements.
- Manage work methods and continuously monitor the implementation of quality assurance standards while recommending corrective actions when deviations occur.
- Develop method statements for project activities, including risk assessments, job safety, environmental analysis, inspection test plans, and checklists based on project specifications.
- Conduct and document electrical and instrumentation-specific inspections and tests, including loop checks, continuity and insulation resistance testing, torque verification, instrument calibration, and factory acceptance tests (FAT) and site acceptance tests (SAT).

- Ensure functional integrity of electrical and instrumentation systems before energization through proper verification and documentation.
- Oversee inbound material inspections and subcontractor quality compliance by verifying test certificates, calibration records, explosion-proof compliance, and workmanship practices such as cable termination, glanding, ferruling, and cable dressing.
- Analyze inspection data, test results, and non-conformance report trends to identify root causes, support corrective actions, and provide quality insights for process improvement and compliance.
- Attend weekly meetings and technical forums related to project quality management and technical discussions.
- Manage time and resources effectively to meet quality control deadlines and prioritize inspections according to project schedules.
- Collaborate with engineers, technicians, project managers, and cross-functional teams to plan, execute, and monitor project activities effectively.
- Demonstrate strong problem-solving abilities by identifying quality issues and recommending effective corrective and preventive actions.
- Mentor junior team members when required and support knowledge transfer within the team.
- Adhere to confidentiality requirements, including Non-Disclosure Agreements (NDA).
- Follow Standard Operating Procedures (SOPs) and company policies in all quality-related activities.
- Perform additional duties and responsibilities assigned by the superior from time to time.

Internal Communication:

- Provide regular updates to management regarding quality control activities, non-conformances, and corrective actions to support informed decision-making.
- Report progress to the Project Manager and seek guidance when necessary.
- Communicate project-specific quality requirements, standards, and procedures to project managers, engineers, and project team members.
- Collaborate with cross-functional teams to address interdisciplinary quality issues and enhance overall project quality.

External Communication:

- Communicate with clients to demonstrate the organization's commitment to quality and provide updates on quality control processes.
- Address client inquiries or concerns related to project quality to ensure client satisfaction and positive project outcomes.
- Coordinate with subcontractors and suppliers to ensure adherence to established quality standards for materials and workmanship.
- Facilitate communication with internal and external auditors by providing necessary documentation and participating in quality audits.

Reporting Structure and Authority:

- QA/QC Engineers typically report to the Project Manager or another senior management role.

- Regular updates on quality control activities, non-conformances, and corrective actions are communicated to ensure alignment with project goals.
- Collaborate closely with project management, engineering, and construction teams to ensure project quality standards are met.
- Provide insights on the impact of quality performance on overall project success to support strategic decision-making.
- May have authority to approve expenditures related to corrective actions required to address non-conformances within specified limits.

Job Requirements:

- Bachelor's degree in Electrical Engineering, Instrumentation Engineering, Electronics Engineering, or a related discipline.
- Minimum of **4 years** of experience in **QA/QC within the water treatment, construction, or oil and gas industries.**
- Strong knowledge of quality assurance and quality control principles, inspection procedures, and testing standards.
- Proficiency in Microsoft Office applications for documentation, reporting, and analysis.
- Ability to read and interpret P&ID drawings, wiring diagrams, and electrical schematics.
- Understanding of SCADA, DCS, and PLC interfaces and related electrical systems.
- Experience in documentation, reporting, and quality governance processes.
- Strong inspection and testing competency for electrical and instrumentation systems.
- Experience in managing non-conformance reports and implementing corrective actions.

- Familiarity with Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), and quality audits.
- Ability to prepare method statements, risk assessments, inspection test plans, and quality documentation.
- Strong analytical and problem-solving skills with the ability to interpret inspection data and test results.
- Knowledge of inbound material inspection and supplier or subcontractor quality control.
- QC-related certifications, such as Certified Quality Engineer (CQE), will be considered an advantage.
- Excellent communication skills in English.
- Flexibility to travel for site deputations within and outside the UAE when required.

Key Skills:

- Comprehensive Quality Assurance and Quality Control expertise.
- Technical knowledge of relevant codes and standards.
- Documentation, reporting, and quality governance.
- Inspection and testing competency.
- NCR management and corrective action implementation.
- FAT and SAT coordination, quality audit, and compliance oversight.
- Method statement preparation and risk assessment.
- Supplier and subcontractor quality control.
- Data analysis and problem-solving abilities.
- Ability to read and interpret P&ID drawings and wiring charts.
- Understanding of SCADA, DCS, and PLC systems.
- MS Office proficiency.